ISM Note 7

Yang Chen

4/9/2026

1. Hydrodynamics (continued)

Last class we ended with the lagrangian derivative, which is given by

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + (v \cdot \nabla)$$

The partial derivative shows the change of the quantity at a fixed point, which is called **the Eulerian description**. While the lagrangian derivative is a total derivative, which is the change of the quantity following the motion of the fluid element.

With that we can introduce the so-called **Archimedes' principle**, which tells us that the heavier fluid will sink, and the lighter fluid will float. This principle can be easily understood from the momentum equation.

We will now introduce the energy equation. Recall that the mass equation is the 0th moment of the Boltzmann equation, and the momentum equation is the 1st moment of the Boltzmann equation, then straightforwardly, we can get the 2nd moment of the Boltzmann equation, which is the so-called energy equation.

However, we will not write down it directly. Instead we will try to write it in the conservation form, which is given by

$$\frac{\partial E}{\partial t} + \nabla \cdot (\frac{1}{2}\rho v^2 + \frac{5}{2}P)\vec{v} = \mathrm{source}$$

Notice that here the flux term contains another pressure. Why for the extra pressure? To understand it, we can decompose it as

$$\nabla \cdot (P\vec{v}) = P(\nabla \cdot \vec{v}) + \vec{v} \cdot \nabla P$$

The first term is **the PdV work**, and the second term is the work done due to **the flow of the fluid**.

I have to remark that the source term in the RHS usually will not vanish, even for the mass equation case. This seems quite bizzard at first glance, but we can understand it with an example. For example, if we consider a bulk of gas around the black hole, the gas can be swallowed by the black hole and this action will act as a **sink term** for the mass equation.

However, for the energy equation, things is a little more complicated. The source term not only contains everything in the momentum equation, but also contains **the heating and cooling terms**.

Before we end this part, I have to mention that there is another equation we can write down, which is the so-called **entropy equation**. Mathematically this equation is derived by substracting the momentum equation from the total energy equation. By doing this, physically we get an equation describing **the change of the internal energy**.

The entropy equation is given by

$$\frac{3}{2}\frac{\mathrm{d}\ln P}{\mathrm{d}t} - \frac{5}{2}\frac{\mathrm{d}\ln\rho}{\mathrm{d}t} = 0$$

If we define the entropy as

$$S = c_v \ln(\frac{P}{\rho^{5/3}})$$

Then the above equation can be rewritten in a simple form as

$$\frac{dS}{dt} = 0$$

which means **the entropy is conserved** without energy source or sink. The interesting thing is, the gas can have **adiabatic cooling or heating**, but if we look at the entropy equation, we can see that these effects will disappear. So if we find that the entropy is changed, then we can conclude that there must be some non-adiabatic cooling or heating.

Convection

Consider a blob in a stratified atmosphere, if we perturb the blob and let it move up, the thing we care about is whether the blob will fall back or continue to move up. Straightforwardly, this requires us to compare the density drop of the perturbed blob with the surrounding medium.

Inside the blob, we know the entropy is conserved, and we assume the blob is in pressure equilibrium with the surrounding medium, so we have

$$P_{\mathrm{in}} = P_{z+\delta z}$$

and

$$S_{\mathrm{in}} = S_{\mathrm{z}}$$

Outside the blob, we have

$$P_{\mathrm{out}} = P_{z+\delta z}$$

and

$$S_{\mathrm{out}} = S_{z+\delta z}$$

So with some simple derivations, we can get

$$\frac{\partial S}{\partial z} > 0$$

acts as **the stability criterion for the convection**. If the blob is stable, it will oscillate aroung the equilibrium position, and the oscillation frequency is called **Brunt-Vaisala frequency**, which is given by

$$N^2 = \frac{g}{c_p}\frac{\partial S}{\partial z}$$

where g is the gravity, and c_p is the specific heat at constant pressure.

As an example, the internal gravity waves in the atmosphere of the Earth is formed due to convection, which is also called G-modes.

Sound waves

We can also use the linear perturbation theory to analyze the sound waves. We assume the background is static, and we have

$$\begin{cases} \rho = \rho_0 + \delta\rho \\ P = P_0 + \delta P \\ v = 0 + \delta v \end{cases}$$

We can plug these into the mass equation and get

$$\frac{\partial}{\partial t}(\delta\rho) + \rho(\nabla \cdot \delta\vec{v}) = 0$$

Then we can do the Fourier transformation and assume

$$\delta \approx e^{i(\vec{k}\cdot\vec{x} - \omega t)}$$

Thus we can get

$$-i\omega\,\delta\rho + i\rho_0\,(\vec{k}\cdot\delta\vec{v}) = 0$$

This equation tells us that **the longitudinal component of the velocity is related to the density perturbation**, and this component is equivalent to the compressive mode.

If $\vec{k}\cdot\delta\vec{v} = 0$, e.g. the wave is solenoidal, there will be no density fluctuations.

For the momentum equation, with the similar procedure, we can get

$$-i\omega\,\rho_0\,\delta\vec{v} + i\vec{k}\,\delta P = 0$$

We dotting $\vec{k}$ on both sides, and then can get

$$-i\omega\rho_0\vec{k}\cdot\delta\vec{v} + ik^2\delta P = 0$$

Then we can plug in the result to the entropy equation, and finally we can get the dispersion relation for the sound wave, which is given by

$$\omega^2 = c_s^2 k^2$$

where c_s is the sound speed, which is given by

$$c_s^2 = \frac{\gamma P_0}{\rho_0}$$

So for a sound wave we know its group velocity is the same as its phase velocity, and it is a **longitudinal wave**.

Convection (WKB)

We write again the momentum equation with a gravity source term

$$-i\omega\rho_0\,\delta\vec{v} + i\vec{k}\,\delta P = \delta\rho\,\vec{g}$$

Cross this equation with $\vec{k}$ twice, and assume $\delta P = 0$, in that case, we can get

$$i\omega c_p \frac{\delta\rho}{\rho} + \delta v_z \frac{dS}{dz} = 0$$

this tells us that **the density will only response to vertical motion**. Then we can derive the dispersion relation for convection, which is given by

$$\omega^2 = N^2(1 - \bar{k}_z^2)$$

where N is the Brunt-Vaisala frequency we defined above, and

$$\bar{k}_z = \frac{k_z}{k}$$

So if $k_z = 0$, and $k_h \neq 0$, then we can get $\delta v \neq 0$.

While if $k_z \neq 0$, and $k_h = 0$, then we can get $\delta v_z = 0$, which means it cannot propagate vertically. So this is a **transverse wave**, and the wave package energy travels along the surface of constant phase. This is what we call **the internal gravity wave**.